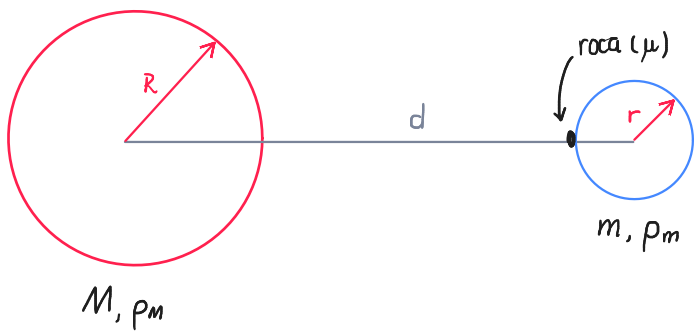




$$F_{a,M} = \frac{GM\mu}{x^2} = a_m(x) \cdot \mu \quad \text{con} \quad a_m(x) = \frac{GM}{x^2}$$

$$\Delta a = a_m(d-r) - a_m(d)$$

$$= GM \left[\frac{1}{(d-r)^2} - \frac{1}{d^2} \right]$$

$$= \frac{GM}{d^2} \left[\frac{1}{(1-r/d)^2} - 1 \right] \quad \text{como } \frac{r}{d} \ll 1, \quad \left(1 - \frac{r}{d}\right)^{-2} \approx \left(1 + 2\frac{r}{d}\right)$$

$$\rightarrow \Delta a = \frac{2GMr}{d^3} \quad (\text{aceleración de marea})$$

$$F_{\text{marea}} = \mu \Delta a = \frac{2GM\mu r}{d^3} \quad \text{y} \quad F_{a,m} = \frac{Gm\mu}{r^2}$$

$$\rightarrow \frac{2GM\mu r}{d^3} = \frac{Gm\mu}{r^2}$$

$$d^3 = \frac{2Mr^3}{m} \quad \text{ahora,} \quad M = \frac{4}{3}\pi R^3 \rho_M \quad \text{y} \quad m = \frac{4}{3}\pi r^3 \rho_m$$

$$d^3 = 2r^3 \left(\frac{M}{m} \right) = 2r^3 \frac{R^3 \rho_M}{r^3 \rho_m}$$

$$\rightarrow \boxed{d = R \left(2 \frac{\rho_M}{\rho_m} \right)^{1/3}}$$